

# JEE ADVANCED WEEKEND TEST SERIES

**44**  
YEARS  
OF EXCELLENCE



**NARAYANA**  
IIT-JEE ACADEMY-INDIA

Sec: JR.IIT\_\*CO-SC(MODEL-A)

Time: 3 Hrs

Date: 04-02-24

Max. Marks: 186

## 04-02-24\_JR.IIT\_STAR CO-SC(MODEL-A)\_JEE ADV\_WAT-38\_SYLLABUS

- MATHEMATICS:** **PRESENT WEEK (80%):** **CIRCLE:** Equation of Normal and its Properties, Equation of Chord whose midpoint is given, Co-Normal points, Common Tangent to Circle and Parabola / Two Parabolas  
**PREVIOUS WEEK(20%):** **Parabola:** Equation of Parabola, Equation of Parabola (Standard form, Foci, Directrix, Eccentricity, Parametric form), Tangent and its Properties, Chord of contact
- PHYSICS:** **PRESENT WEEK (80%):** Introduction to dielectrics, discussion on effect of dielectric when introduced in to a capacitor (isolated or in a circuit). Energy related and switching problems of capacitors involving dielectrics (**Exclude: RC Circuits**)  
**PREVIOUS WEEK (20%):** **CAPACITORS:** Introduction, types of capacitors, combination of capacitors in series and in parallel, general capacitor circuits without involvement of dielectric, Energy stored in a capacitor (without dielectric), charge flown and heat generated problems when switch is thrown from one position to other (**Exclude: Dielectric**)
- CHEMISTRY:** **PRESENT WEEK (80%):** **Alcohols:** Preparation of alcohols from alkene, alkyl halides, amines, aldehydes, ketones and carboxylic acid and derivatives, Physical properties, reactions: esterification, dehydration, with metals, conversion into alkyl chlorides, oxidation into aldehyde, ketones and carboxylic acids, Lucas test, Victor Meyer test (**Exclude: MPV reduction, exceptions of LAH & SBH, Swern oxidation**)  
**Glycol and glycerol:** Preparation, reaction with conc. HI, PI<sub>3</sub>, HIO<sub>4</sub>, glycerol with oxalic acid, pinacol-pinacolone rearrangement  
**PREVIOUS WEEK (20%):** **Aryl halides:** preparation using EAS, Sandmeyer reaction, Mechanism of S<sub>N</sub>Ar reactions (**Exclude: Benzyne mechanism & cine substitution**) and EAS Reactions chlorobenzene, Fittig, Wurtz-Fittig **Grignard reagent:** Preparation and properties of Grignard reagent & Organo lithium

**MATHEMATICS :**

| Section                  | Question Type                                                            | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|--------------------------|--------------------------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 1 – 5)     | Questions with Single Correct Choice                                     | 3         | -1         | 5         | 15          |
| Sec – II(Q.N : 6 – 13)   | Questions with Multiple Correct Choice<br>(Partial Marking +1)           | 4         | -2         | 8         | 32          |
| Sec – III(Q.N : 14 – 18) | Questions with Integer Answer Type<br>(Single digit integer 0 to 9 Only) | 3         | 0          | 5         | 15          |
| <b>Total</b>             |                                                                          |           |            | <b>18</b> | <b>62</b>   |

**PHYSICS :**

| Section                  | Question Type                                                            | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|--------------------------|--------------------------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 19 – 23)   | Questions with Single Correct Choice                                     | 3         | -1         | 5         | 15          |
| Sec – II(Q.N : 24 – 31)  | Questions with Multiple Correct Choice<br>(Partial Marking +1)           | 4         | -2         | 8         | 32          |
| Sec – III(Q.N : 32 – 36) | Questions with Integer Answer Type<br>(Single digit integer 0 to 9 Only) | 3         | 0          | 5         | 15          |
| <b>Total</b>             |                                                                          |           |            | <b>18</b> | <b>62</b>   |

**CHEMISTRY :**

| Section                  | Question Type                                                            | +Ve Marks | - Ve Marks | No.of Qs  | Total marks |
|--------------------------|--------------------------------------------------------------------------|-----------|------------|-----------|-------------|
| Sec – I(Q.N : 37 – 41)   | Questions with Single Correct Choice                                     | 3         | -1         | 5         | 15          |
| Sec – II(Q.N : 42 – 49)  | Questions with Multiple Correct Choice<br>(Partial Marking +1)           | 4         | -2         | 8         | 32          |
| Sec – III(Q.N : 50 – 54) | Questions with Integer Answer Type<br>(Single digit integer 0 to 9 Only) | 3         | 0          | 5         | 15          |
| <b>Total</b>             |                                                                          |           |            | <b>18</b> | <b>62</b>   |

## MATHEMATICS

Max Marks: 62

SECTION – I  
(SINGLE CORRECT ANSWER TYPE)

This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.

1. Let AB be a chord of the parabola  $x^2 = 4y$ . A circle drawn with AB as diameter passes through the vertex C of the parabola. If area of  $\triangle ABC = 20$  sq units then the coordinates of A can be  
A) (6,9)                      B) (-6,9)                      C) (-8,16)                      D) (8,-16)
2. If the minimum distance between the parabola  $y^2 - 4x - 8y + 40 = 0$  and  $x^2 - 8x - 4y + 40 = 0$  is d. Then  $d^2$  is  
A) 1                      B) 0                      C) 3                      D) 2
3. Let  $P: y^2 = 4ax$ ,  $a > 0$  be a parabola with focus S. let the tangents to the parabola P make angle of  $\frac{\pi}{4}$  with the line  $y = 3x + 5$  touch the parabola P at A and B, Then the value of a for which A,B and S are collinear is  
A) 8 only                      B) 2 only                      C)  $\frac{1}{4}$  only                      D) Any  $a > 0$
4. A variable chord PQ of parabola  $y^2 = 4x$  is drawn parallel to a line  $y = x$ . Then which of the following point lies on the locus of point of intersection of the normals at P and Q  
A) (7,2)                      B) (7,1)                      C) (3,6)                      D) (10,2)
5. If the straight line  $3x + 4y + 1 = 0$  is a normal to the parabola  $y^2 + \lambda x = 0$ , then the distance between focus and directrix of the parabola is  
A) 64/123                      B) 32/123                      C) 16/123                      D) 80/123

**SECTION – II**  
**(MULTIPLE CORRECT ANSWER TYPE)**

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

**Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases**

6. Consider the parabola  $y^2 = 4x$ . A chord PQ is normal to the parabola at  $P(4, 4)$ ,  
**and**  $\angle PSQ = 90^\circ$  where S is its focus. Tangents drawn to the parabola  $y^2 = 4x$  at P and Q intersect at T. R is the midpoint of TQ then
- A)  $SQ = 2SP$
- B)  $\angle TSR = 90^\circ$
- C)  $ST = 2SR$
- D) Length of the common chord of circumcircles of the triangles SPQ, SQR is 10
7. Let P be a point on the parabola  $y^2 = 4x$  such that normal chord drawn at P to parabola has shortest length. If tangent and normal at P intersect axis of parabola at A and B respectively then which of the following is/are CORRECT?
- A) Slope of normal at P is  $\pm 2$
- B) Slope of tangent at P is  $\pm \frac{1}{\sqrt{2}}$
- C) Area of  $\triangle PAB$  is  $12\sqrt{2}$  sq. units
- D) **Circumcentre** of  $\triangle PAB$  is  $(1, 0)$
8. A straight line  $y = m(x - a)$ ,  $m \neq 0$  meets a parabola  $y^2 = 4ax$  at two points A and B. Then which of the following option(s) is/are correct?
- A) Length  $AB = 4a$  when  $m = 1$
- B) Length  $AB = 8a$  when  $m = 1$
- C) Angle between normals drawn at A and B is  $45^\circ$  when  $m = 1$
- D) Angle between normals drawn at A and B is  $90^\circ$  when  $m = 2$
9. A circle cuts the parabola  $x^2 = 16y$  at  $O(0, 0)$ ,  $A(-4, 1)$ ,  $B(x_1, y_1)$ ;  $x_1 < 0$ ; and  $C(x_2, y_2)$ . If normals at A and B to the parabola intersect at C, then
- A)  $x_1 + y_2 = 49$
- B)  $x_2 + y_1 = 100$
- C)  $(0, -8)$ , A **and** B are collinear
- D)  $(0, 8)$ , A **and** C are collinear



10. A line  $y = 2x + c$  intersects the parabola  $y^2 = 4x$  in two distinct points A and B, such that normals at A and B intersect on the principal axis of parabola. Then
- A) AB is perpendicular to principal axis
  - B) Exactly one of A&B is vertex of parabola
  - C) AB subtends right angle at focus
  - D) Foot of perpendicular from focus upon AB divides AB in the ratio 1:2 or 2:1
11. TP and TQ are tangents to parabola  $y^2 = 4x$  and normals at P and Q intersect at point R on the same curve. The locus of the centre of the circle circumscribing  $\triangle TPQ$  is a parabola whose
- A) Vertex is (1,0)
  - B) Foot of directrix is  $\left(\frac{7}{8}, 0\right)$
  - C) Length of latus rectum is  $\frac{1}{4}$
  - D) Focus is  $\left(\frac{9}{8}, 0\right)$
12. The locus of the point of intersection of those normals to the parabola  $x^2 = 8y$  which are at right angles to each other, lies on a parabola then. Which of the following hold(s) good in respect of second parabola?
- A) Length of the latus rectum is 2
  - B) Coordinates of focus are  $\left(0, \frac{11}{2}\right)$
  - C) Equation of a director circle is  $2y - 11 = 0$
  - D) Equation of axis of symmetry is  $y = 0$
13. If the two parabolas  $y^2 = 4x$  and  $y^2 = (x - k)$  have a common normal other than  $x$ -axis, then value of k can be:
- A) 1
  - B) 2
  - C) 3
  - D) 4

**SECTION – III**  
**(INTEGER ANSWER TYPE)**

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

**Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases**

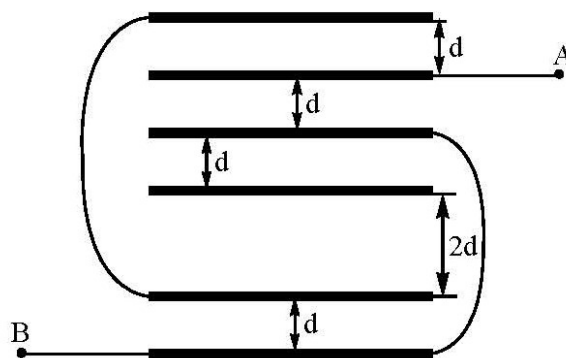
14. A quadrilateral ABCD is inscribed in a parabola having equation  $y^2 = 16x$  such that three of its side AB, BC and CD passes through the fixed points  $(-2, 0)$ ,  $(3, 0)$  and  $(-6, 0)$  respectively. Then the fourth side DA will pass through a fixed point  $(h, 0)$  where  $h$  is equal to
15. A circle having its centre at  $(2, 3)$  is cut orthogonally by the parabola  $y^2 = 4x$ . If  $(x_1, y_1)$  and  $(x_2, y_2)$  ( $x_2 > x_1$ ) are possible points of intersection of the two curves then the value of  $x_2 + 2y_2 - 2x_1 - y_1$  is
16. From a variable point  $P(t^2, 2t)$  ( $1 \leq t \leq 3$ ) on the parabola  $y^2 = 4x$ , perpendicular PM is drawn to the tangent at the vertex of the parabola. Now from the midpoint Q of PM, perpendicular QL is drawn to the focal chord of parabola through P. If maximum length of QL is  $\lambda$  then value of  $[\lambda]$  is \_\_\_\_ ( $[\cdot]$  represents greatest integral function)
17. Let a parabola  $y^2 = \sqrt{2}x$ . From a point where its directrix intersects the line  $2\sqrt{2}x - \sqrt{2}y + 3 = 0$ , a normal to the parabola is drawn, meeting the parabola at M (foot of normal) and N. R is point of intersection of tangent at M and chord ON (produced, O is origin). If  $\angle MRN$  is  $\theta$  then value of  $|\tan \theta|$  is
18. Let  $P(a, b)$  and  $Q(c, d)$  are the two points on the parabola  $y^2 = 8x$  such that the normals at them meet in  $(18, 12)$ . If the product of  $(abcd)$  is N, then sum of digits of N is \_\_\_\_.

**SECTION – I**  
**(SINGLE CORRECT ANSWER TYPE)**

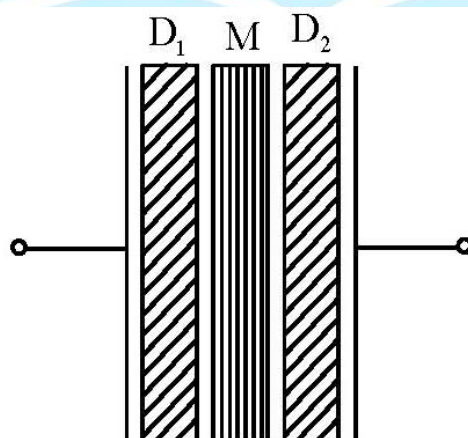
This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.**

19. What is the equivalent capacitance between points A and B. Each plate has area A (one side) and negligible thickness. Given  $C = \frac{\epsilon_0 A}{d} = 7 \mu F$

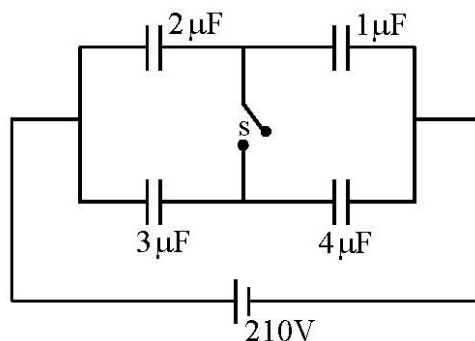


- A)  $7 \mu F$       B)  $11 \mu F$       C)  $12 \mu F$       D)  $15 \mu F$
20. Plates of a parallel plate capacitor have area A and distance between them is d. A cell of e.m.f V is connected across the capacitor. Two identical dielectric slabs ( $D_1$  and  $D_2$ ) and a metal slab (M), each of thickness  $d/3$  and area A, are inserted in the capacitor as shown. Relative permittivity of each of the dielectric slab is  $\epsilon_r$ . What is the modulus of the net force on either of the capacitor plates?



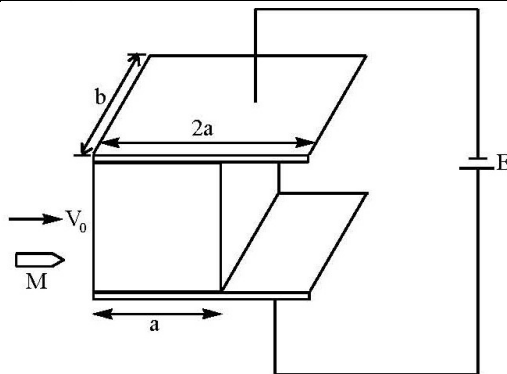
- A)  $\frac{1}{2} \epsilon_0 A \left( \frac{3\epsilon_r V}{4d} \right)^2$       B)  $\frac{1}{2} \epsilon_0 A \left( \frac{5\epsilon_r V}{2d} \right)^2$       C)  $\frac{1}{2} \epsilon_0 A \left( \frac{3\epsilon_r V}{2d} \right)^2$       D)  $\frac{1}{2} \epsilon_0 \epsilon_r A \left( \frac{3V}{2d} \right)^2$

21. Consider a network shown in the figure. Initially the switch S was open. If the switch S is closed now, then which of the following options are correct?



- A) The amount of charge flowing through the switch is  $105 \mu C$   
 B) The amount of charge flowing through the switch is  $35 \mu C$   
 C) The amount of charge flowing through the switch is  $70 \mu C$   
 D) The amount of charge flowing through the switch is  $140 \mu C$
22. There is a parallel-plate capacitor of capacitance  $20 \text{ nF}$  (when there is nothing between the plates). A thick metallic plate having the same length and breadth as that of the capacitor plates but with a thickness of  $0.6$  times the gap between the plates of the capacitor, is now inserted parallel to the plates without touching them and a battery of  $100 \text{ V}$  is connected across the capacitor. What is the work done in slowly pulling the metallic plate out of the capacitor?
- A)  $0.05 \text{ mJ}$       B)  $0.10 \text{ mJ}$       C)  $0.15 \text{ mJ}$       D)  $0.20 \text{ mJ}$
23. A parallel plate capacitor is half filled with a solid dielectric of relative permittivity ' $K$ ' and mass ' $M$ '. A cell of voltage  $E$  is connected across the capacitor. The plates of the capacitor are smooth and fixed. Distance between plates is  $d$ . A bullet of mass ' $M$ ' hits the dielectric and gets embedded in it. It is found that the dielectric just exits the capacitor completely. The speed of the bullet is





A)  $E \left[ \frac{2\varepsilon_0 ab(K-1)}{Md} \right]^{1/2}$

B)  $E \left[ \frac{\varepsilon_0 ab(K-1)}{Md} \right]^{1/2}$

C)  $2E \left[ \frac{\varepsilon_0 ab(K-1)}{Md} \right]^{1/2}$

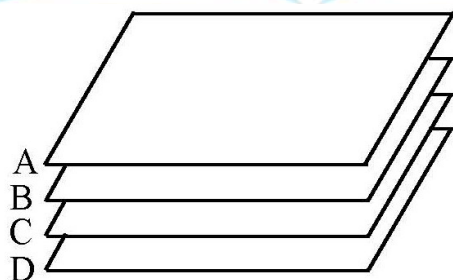
D)  $E \left[ \frac{\varepsilon_0 ab(K-1)}{2Md} \right]^{1/2}$

## SECTION – II (MULTIPLE CORRECT ANSWER TYPE)

This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

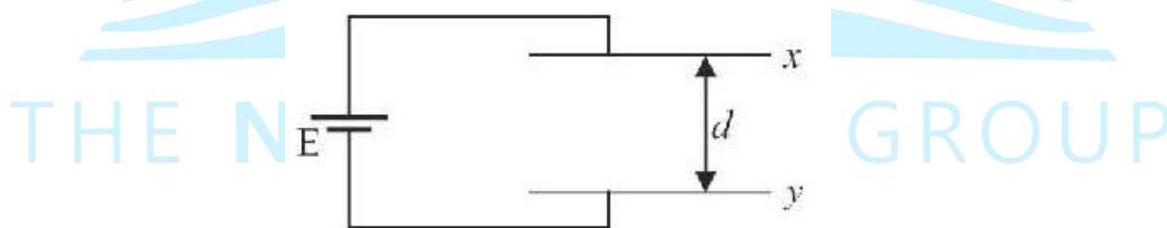
**Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases**

24. Four identical **thin** metallic plates are arranged very close and parallel to each other as shown in the figure. Plates A and D are connected across a voltage source, which maintains potential of plate A higher than that of D by an amount  $V_0$ . Plates B and C are connected by a short conducting **wire**. Now the voltage source is removed, **then the wire connecting B and C is also removed**. Now plates A and D are connected by another short conducting wire. **Identify the correct statement(s) of the following**



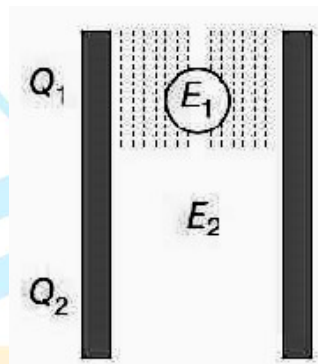
- A) Potential difference between A and B has magnitude  $V_0 / 6$ , with A at higher potential
- B) Potential difference  $V_C - V_A$  is equal to potential difference  $V_D - V_B$
- C) The potential of B is higher than that of D
- D) Potential of C is  $V_0 / 3$  higher than the potential of B

25. A parallel plate capacitor of capacitance  $10\ \mu F$  is connected to a cell of emf 10 volt and fully charged. Now a dielectric slab ( $k = 3$ ) of thickness equal to the gap between the plates is very slowly inserted **by someone** to completely fill in the gap, keeping the cell connected. During the filling process. **(consider no electrical resistance anywhere)**
- A) the increase in charge on the capacitor  $200\ \mu C$
- B) the heat produced **in the “capacitor + cell” system is zero**
- C) Energy supplied by the cell = increase in stored potential energy + work done on the person who is filling the dielectric slab.
- D) the increase in charge on the capacitor  $400\ \mu C$
26. A parallel plate air capacitor having separation 'a' between the plates is charged to Q and battery is disconnected from the capacitor. Now the separation between the plates is increased to '2a' slowly by external force. The work done by external force is numerically equal to:
- A) Electrostatic energy stored in the increased volume of the capacitor
- B) Half of the electrostatic energy stored in the increased volume of the capacitor
- C) Two times the electrostatic energy stored in the increased volume of the capacitor
- D) electrostatic energy stored in the initial volume of the capacitor
27. A parallel-plate capacitor has plate area A, the distance between its two plates x and y is d and a dielectric slab of dielectric constant  $\epsilon_r$  is filled between the regions. It **is connected** to a cell of emf E. If the dielectric slab is removed, then

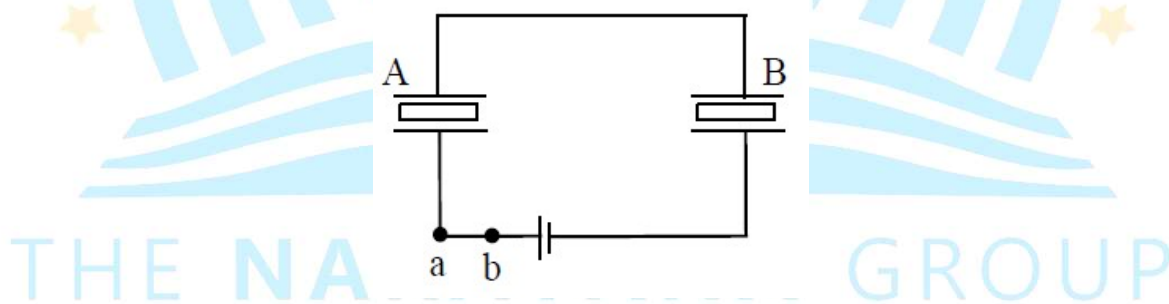


- A) Battery does negative work
- B)  $\frac{\epsilon_0 A}{d} [\epsilon_r - 1] E$  charge flows through the circuit
- C) negative charge flows from plate y to plate x
- D) negative charge flows from plate x to plate y

28. A parallel plate capacitor has a dielectric slab of dielectric constant  $K$  between its plates that covers  $1/3$  of the area of its plates, as shown in the figure. The total capacitance of the capacitor is  $C$  while that of the portion with dielectric in between is  $C_1$ . When the capacitor is charged, the plate area covered by the dielectric gets charge  $Q_1$  and the rest of the area gets charge  $Q_2$ . The electric field in the dielectric is  $E_1$  and that in the other portion is  $E_2$ . Choose the correct option/options, ignoring edge effects.

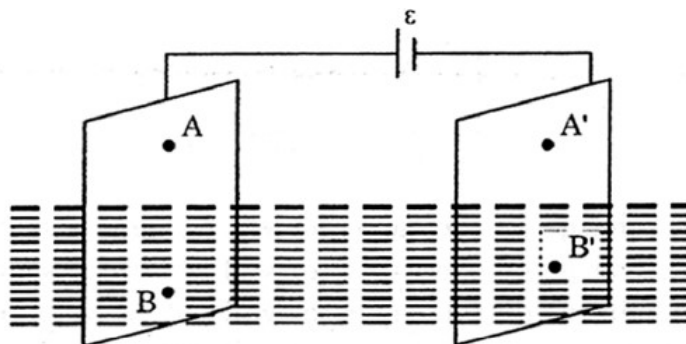


- A)  $\frac{E_1}{E_2} = 1$       B)  $\frac{E_1}{E_2} = \frac{1}{K}$       C)  $\frac{Q_1}{Q_2} = \frac{3}{K}$       D)  $\frac{C}{C_1} = \frac{2+K}{K}$
29. Identical dielectric slabs are inserted into two identical capacitors A and B. These capacitors and a chargeable battery are connected as shown in the figure. Now the slab of capacitor B is pulled out **completely** with battery remaining connected:

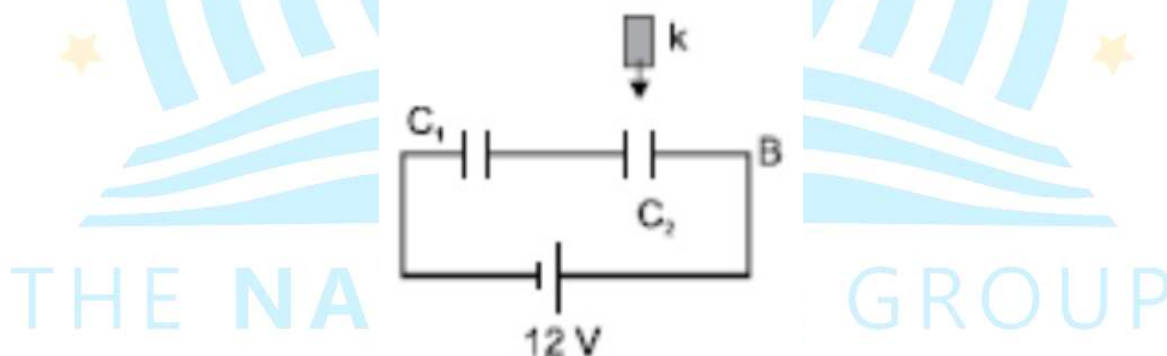


- A) During the process, positive charge flows from  $a$  to  $b$
- B) Finally, charge on capacitor B will be less than that on A
- C) During the process, positive work is done by the external force  $F$ , part of which appears as heat in the circuit.
- D) During the process, internal energy of the battery increases

30. A parallel plate capacitor is dipped in a dielectric liquid vertically, so that half of capacitor is in liquid. Plates of capacitor are joined to an ideal battery of emf  $\varepsilon$ . A, B, A' and B' are four points on **plates on facing surfaces(towards each other)** as shown. Which of the following is/are correct?



- A)  $V_A - V_{B'} = V_A - V_{A'}$   
 B)  $V_A - V_{B'} = V_B - V_{A'}$   
 C) Electric field around A in air = Electric field around B' in liquid.  
 D) Electric field around B in liquid = Electric field around B' in liquid.
31. Two capacitors  $C_1 = 8\mu\text{F}$  and  $C_2 = 4\mu\text{F}$  are connected in series between points A and B. An ideal cell of emf 12 V is connected between A and B for a long time. Now a slab of dielectric constant  $k = 2$  is slowly and fully filled in the gap of capacitor  $C_2$ .



- A) During the filling process the energy taken from the cell is  $192\mu\text{J}$   
 B) In the filling process the increase in electrostatic energy stored in the capacitors is  $96\mu\text{J}$   
 C) The work done by dielectric slab on the filling agent during the filling process is positive.  
 D) The work done by dielectric slab on the filling agent during the filling process is zero.



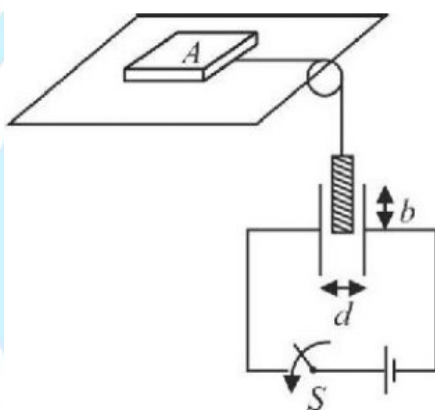
### SECTION – III (INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

**Marking scheme +3 for correct answer , 0 if not attempted and 0 in all other cases**

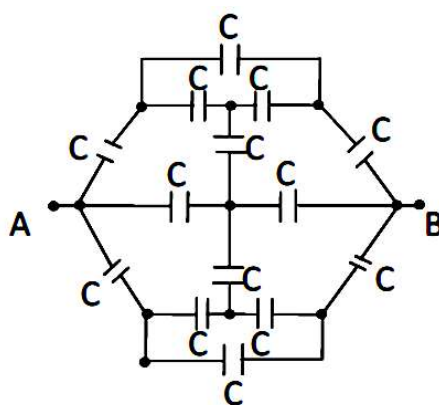
32. A block  $A$  of mass  $M$  kept on a rough horizontal surface is connected to a dielectric slab of mass  $M/6$  and dielectric constant  $k$  by means of a light and inextensible string passing over a fixed pulley as shown in the figure. The dielectric can completely fill the space between the parallel plate capacitor of plate area  $\ell \times \ell$  and with plates separation  $d$  ( $d \ll \ell$ ) kept in vertical position. Initially switch  $S$  is open and length of the dielectric inside the capacitor is  $b$  and the capacitor is chargeless. The coefficient of friction between the block  $A$  and the surface is  $\mu = \frac{1}{4}$ . Ignore any other friction.

(Given  $\frac{Mgd}{\epsilon_0 \ell (k-1)} = 24$ )

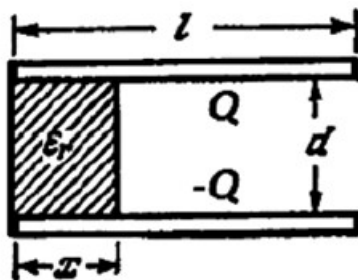


Find the minimum value of the emf  $V$  of the battery so that after closing the switch the block  $A$  will move.

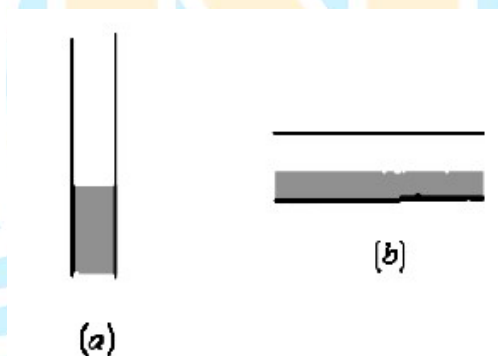
33. In the following figure all capacitors are of capacitance  $C$ . The equivalent capacitance across  $A$  and  $B$  is  $\frac{x C}{4}$  where  $x$  is an integer. Find  $x$ .



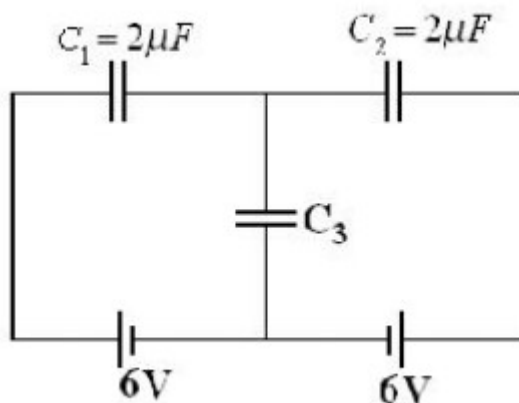
34. Two conducting square plates of side length  $l$  are arranged parallel to each other at a distance  $d$  ( $d \ll l$ ). The space between the plates is filled with a dielectric of relative permittivity  $\epsilon_r = 4$  up to a length  $x = l/3$ . The upper plate is given a charge  $Q$  and the lower plate is given a charge  $-Q$  as shown in the figure. If  $Q = 6\mu C$  then find how much bound charge (in  $\mu C$ ) is present on the lower surface of dielectric?



35. A vertical parallel – plate capacitor is half filled with a dielectric for which the dielectric constant is 2. When this capacitor is positioned horizontally, the fraction of it to be filled with the same dielectric in order that the capacitor to have equal capacitance in both cases is  $\frac{n}{m}$ . Find the value of (n.m).



36. Three uncharged capacitors are connected in the circuit as shown. What should be the value of  $C_3$  (in  $\mu F$ ) so that potential difference across each capacitor is same.



## CHEMISTRY

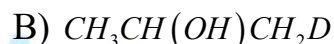
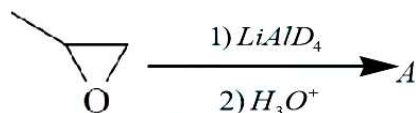
Max Marks: 62

**SECTION – I**  
**(SINGLE CORRECT ANSWER TYPE)**

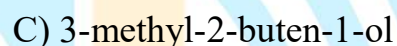
This section contains 5 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONLY ONE option can be correct.

**Marking scheme: +3 for correct answer, 0 if not attempted and –1 in all other cases.**

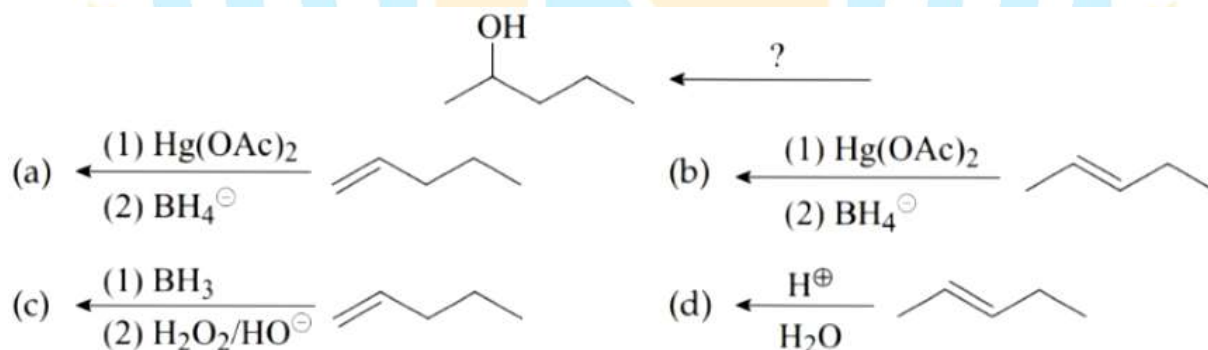
37. Identify the predominant compound A in the following reaction:



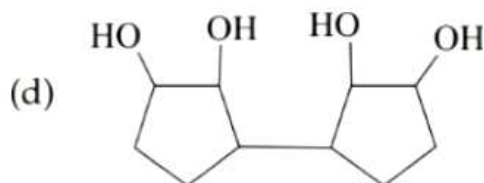
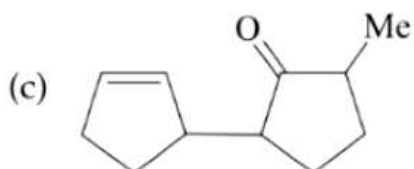
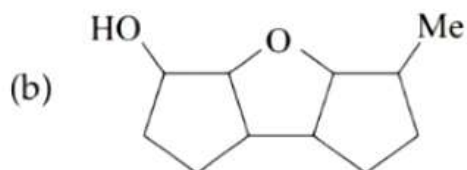
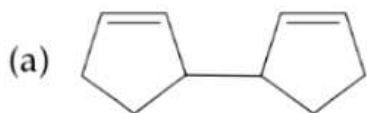
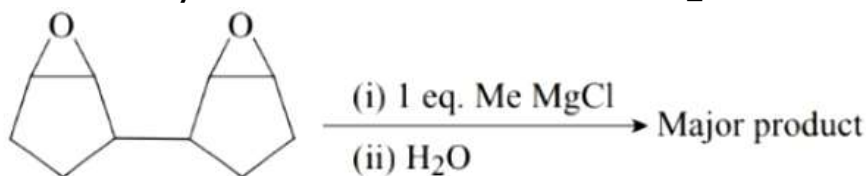
38. A chiral  $\text{C}_5\text{H}_{10}\text{O}$  alcohol is reduced by catalytic hydrogenation to an achiral  $\text{C}_5\text{H}_{12}\text{O}$  alcohol. The original alcohol is oxidized by activated  $\text{MnO}_2$  to an achiral carbonyl compound ( $\text{C}_5\text{H}_8\text{O}$ ) which of the following might be the chiral alcohol?



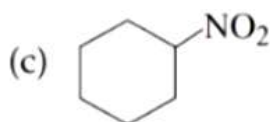
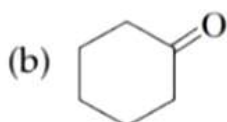
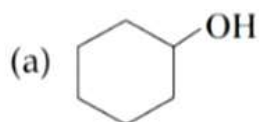
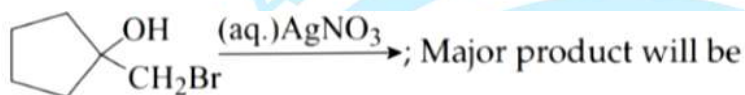
39. Which is the best synthesis of the following alcohol?



40.



41.

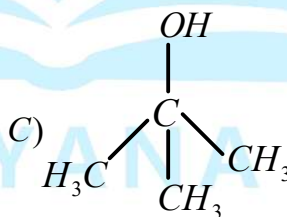
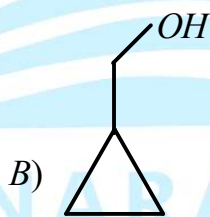
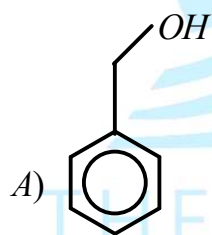


## SECTION – II (MULTIPLE CORRECT ANSWER TYPE)

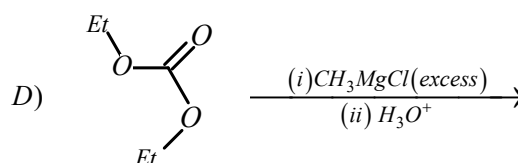
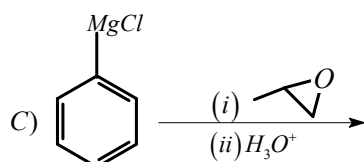
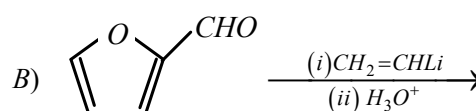
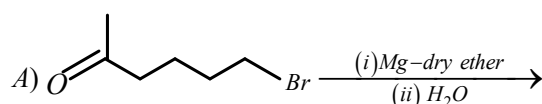
This section contains 8 multiple choice questions. Each question has 4 options (A), (B), (C) and (D) for its answer, out of which ONE OR MORE than ONE option can be correct.

**Marking scheme: +4 for all correct options & +1 partial marks, 0 if not attempted and -2 in all wrong cases**

42. Which of the following will produce immediate turbidity with Lucas reagent?



43. In which of the following reactions 3° alcohol is not formed?





44. How many of the following on hydride reduction (i.e. with  $LiAlH_4$  or  $NaBH_4$ ) give a mixture of diastereometric alcohols?

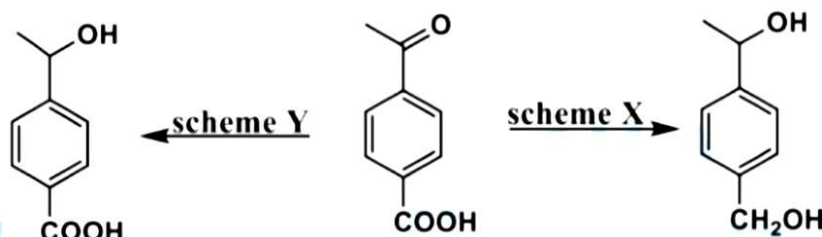
A) (S)-3-phenyl-2-butanone

B) 4-tert-butylcyclohexanone

C) p-methylbenzaldehyde

D) (R)-2-methylcyclopentanone

45. Consider the following reaction schemes. The correct set of reagents to be used in scheme A and/or scheme B is/are



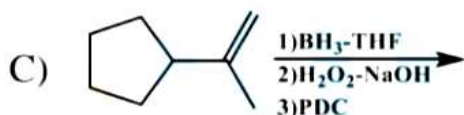
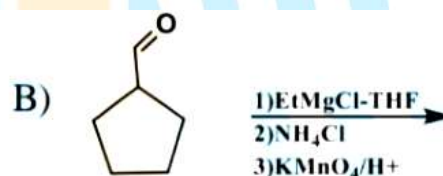
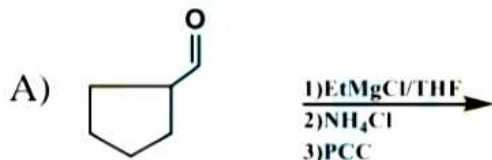
A) Scheme X :  $LiAlH_4 - Et_2O / H_2O$

B) Scheme Y :  $NaBH_4 / EtOH$

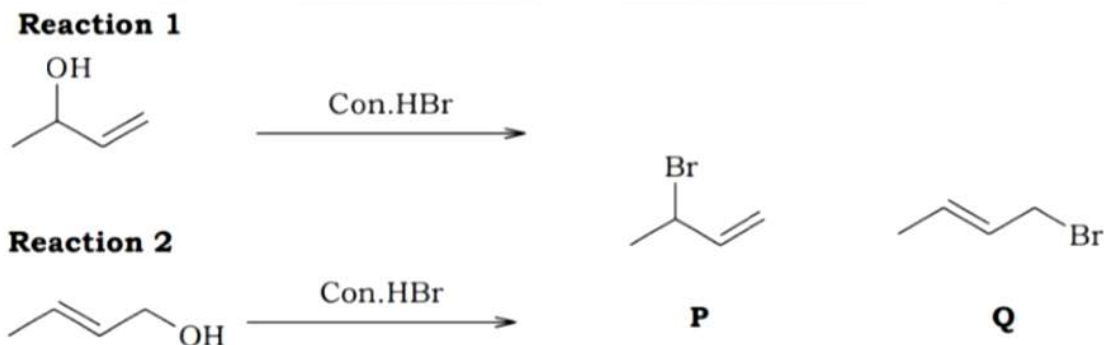
C) Scheme Y :  $LiAlH_4 - Et_2O / H_2O$

D) Scheme X :  $BH_3 - THF$

46. Which of the following reaction(s) result(s) in carbonyl compound as product(s)



47. Correct statement about the product(s) of the following reaction is



A) P is major product in Reaction 1 while Q is major product in Reaction 2

B) Q is major product in Reaction 1 while P is major product in Reaction 2

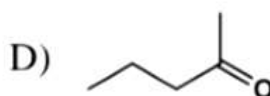
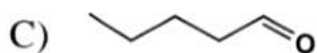
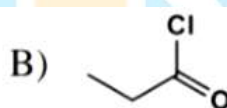
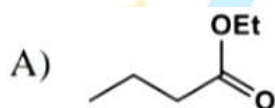
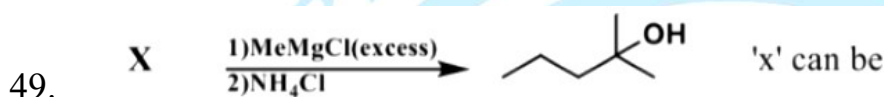
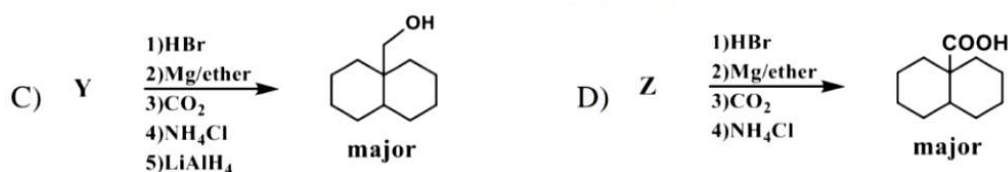
C) Q is major product and P is minor product in both Reaction 1 as well as Reaction 2

D) Both the reactions follow SN1 mechanism in forming one of their products

48. Choose the correct statement(s) regarding given reaction scheme.



A) 'X' can liberate  $H_2$  gas with Na metal    B) Degree of Unsaturation of 'Y' is 3



### SECTION – III (INTEGER ANSWER TYPE)

This section contains 5 questions. The answer is a single digit integer ranging from 0 to 9 (both inclusive).

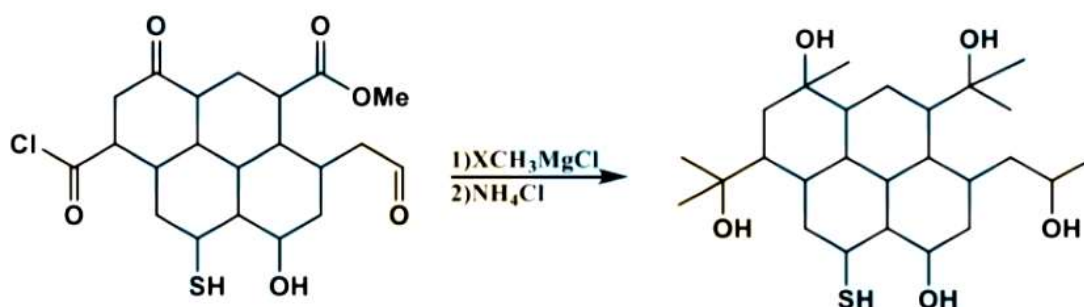
**Marking scheme +3 for correct answer, 0 if not attempted and 0 in all other cases**

50. Action of Grignard reagent on carbonyl compound followed by hydrolysis gives alcohols. In the preparation of all isomeric (structural) butyl alcohols, the number of different carbonyl compounds involved is 'X' and Grignard reagents is 'Y'. Then the value of  $X + Y$  is?

51. The number of moles of  $LiAlH_4$  required to reduce eight moles of carboxylic acid to primary alcohol  $RCOOH \xrightarrow[2) H_3O^+]{1) LiAlH_4} RCH_2OH$

52. During the conversion of one mole of Glycerol in to isopropyl iodide number of moles of HI molecules consumed is?

53. The value of 'X' in the following reaction is (X = no of equivalents of  $\text{CH}_3\text{MgCl}$ )



54. 'X' is the number of reactions which produce at least one alcoholic final product, then 'X' is

